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NEW LOWER SEMICONTINUITY RESULTS FOR NONCONVEX FUNCTIONALS DEFINED ON MEASURES

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1. INTRODUCTION

IN SOME recent papers (see [1, 2, 3, 10, 5]) functionals of the form

$$F(\lambda) = \int_\Omega f\left(x, \frac{d\lambda}{d\mu}\right) d\mu + \int_\Omega f^\infty\left(x, \frac{d\lambda^s}{d|\lambda^s|}\right) d|\lambda^s| \tag{1.1}$$

have been studied, where Ω is a locally compact space, μ is a given positive measure on Ω, λ varies in the space $\mathfrak{M}(\Omega; \mathbf{R}^n)$ of all n-vector measures with bounded variation, $f^\infty(x, s)$ is the recession function of $f(x, s)$ with respect to s, and $\lambda = d\lambda/d\mu \cdot \mu + \lambda^s$ is the Lebesgue–Nikodym decomposition of λ into absolutely continuous and singular parts with respect to μ.

For functionals of the form (1.1), the convexity condition is shown to be necessary in order to have the lower semicontinuity in the weak* convergence of $\mathfrak{M}(\Omega; \mathbf{R}^n)$; moreover, it is proved in [2] that every convex and weak* lower semicontinuous mapping F: $\mathfrak{M}(\Omega; \mathbf{R}^n) \to [0, +\infty]$ is representable in the form (1.1) for a suitable convex integrand $f(x, s)$, provided the addivity condition

$$F(\lambda_1 + \lambda_2) = F(\lambda_1) + F(\lambda_2) \qquad \text{whenever } \lambda_1, \lambda_2 \in \mathfrak{M}(\Omega; \mathbf{R}^n) \text{ with } \lambda_1 \perp \lambda_2 \tag{1.2}$$

is satisfied.

When Ω is an open subset of $\mathbf{R}^n$, μ is the Lebesgue measure, and λ is the gradient Du of a function $u \in BV(\Omega)$, the functional (1.1) takes the form

$$\int_\Omega f(x, \nabla u)\, dx + \int_\Omega f^\infty\left(x, \frac{D^s u}{|D^s u|}\right) d|D^s u| \tag{1.3}$$

where ∇u and $D^s u$ respectively denote the absolutely continuous and the singular parts with respect to the Lebesgue measure. In this case the lower semicontinuity results obtained for (1.1) allow to apply the standard direct method of the calculus of variations for proving the existence of solutions of several minimization problems involving (1.3).

However, functionals of the form (1.1) or (1.3) are not appropriated to treat a new class of problems in which a nonconvex term is present. The simplest example is the minimization

problem for the functional

$$\int_\Omega |\nabla u|^2 \, dx + \int_{S_u} dH^{n-1} + \int_\Omega |u - g(x)|^2 \, dx \tag{1.4}$$

which seems to be important in the mechanics of fractures, in the theory of liquid crystals, and in problems of computer vision (we refer to De Giorgi [8] and to De Giorgi and Ambrosio [9] for a general presentation of the problem). Here u is a function in $BV(\Omega)$, $g \in L^2(\Omega)$, S_u is the set of "jumps" that is the set of all points x such that the upper and lower approximate limits $u^+(x)$ and $u^-(x)$ do not coincide, and H^{n-1} is the $(n-1)$ dimensional Hausdorff measure.

In the one-dimensional case, setting $\lambda = Du$, the principal part of functional (1.4) belongs to the class we consider in the present paper: with the same notation of (1.1) this class consists of all functionals on $\mathfrak{M}(\Omega; \mathbf{R}^n)$ of the form

$$\int_\Omega f\left(x \frac{d\lambda}{d\mu}\right) d\mu + \int_{\Omega - A_\lambda} f^\infty\left(x, \frac{d\lambda^s}{d|\lambda^s|}\right) d|\lambda^s| + \int_{A_\lambda} g(x, \lambda(\{x\})) \, d\#(x) \tag{1.5}$$

where A_λ is the set of all atoms of λ, and $\#$ is the counting measure. For this kind of functionals we establish a lower semicontinuity result (theorem 3.3) with respect to the weak* convergence on $\mathfrak{M}(\Omega; \mathbf{R}^n)$.

2. NOTATION AND PRELIMINARY RESULTS

In this section we fix the notation we shall use in the following. Throughout the paper $(\Omega, \mathcal{B}, \mu)$ will denote a measure space, where Ω is a separable locally compact metric space with distance d, $\mathcal{B}$ is the σ-algebra of Borel subsets of Ω, and $\mu: \mathcal{B} \to [0, +\infty[$ is a positive, finite, nonatomic measure. For ever vector-valued measure $\lambda: \mathcal{B} \to \mathbf{R}^n$ and every $B \in \mathcal{B}$ the variation of λ on B is defined by

$$|\lambda|(B) = \sup\left\{ \sum_{n=1}^{\infty} |\lambda(B_h)| : \bigcup_{h=1}^{\infty} B_h \subset B, \, B_h \text{ pairwise disjoint} \right\}. \tag{2.1}$$

In this way, the set function $B \to |\lambda|(B)$ turns out to be a positive measure, which will be denoted by $|\lambda|$.

The following spaces will be considered.

$C_c(\Omega; \mathbf{R}^n)$ the space of all continuous functions $u: \Omega \to \mathbf{R}^n$ with compact support;

$C_0(\Omega; \mathbf{R}^n)$ the space of all continuous functions $u: \Omega \to \mathbf{R}^n$ "vanishing on the boundary", that is for every $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subset \Omega$ such that $|u(x)| < \varepsilon$ for all $x \in \Omega - K_\varepsilon$;

$\mathfrak{M}(\Omega; \mathbf{R}^n)$ the space of all vector-valued measures $\lambda: \mathcal{B} \to \mathbf{R}^n$ with finite variation on Ω.

It is well-known that $\mathfrak{M}(\Omega; \mathbf{R}^n)$ can be identified with the dual space of $C_0(\Omega; \mathbf{R}^n)$ by the duality

$$(\lambda, u)_\Omega = \int_\Omega u \, d\lambda \qquad (u \in C_0(\Omega; \mathbf{R}^n), \, \lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)),$$

and for every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ we have

$$|\lambda|(\Omega) = \sup\{|\langle \lambda, u\rangle_\Omega| : u \in C_0(\Omega; \mathbf{R}^n), \, \|u\|_{C_0(\Omega; \mathbf{R}^n)} \le 1\}. \tag{2.2}$$

The space $\mathfrak{M}(\Omega; \mathbf{R}^n)$ will be endowed with the weak* topology deriving from the duality between $\mathfrak{M}(\Omega; \mathbf{R}^n)$ and $C_0(\Omega; \mathbf{R}^n)$; in particular, a sequence $\{\lambda_h\}$ in $\mathfrak{M}(\Omega; \mathbf{R}^n)$ will be said to w^*-converge to $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ (and this will be indicated by $\lambda_h \to \lambda$) if an only if

$$\langle \lambda_h, u \rangle_\Omega \to \langle \lambda, u \rangle_\Omega \qquad \text{for every } u \in C_0(\Omega; \mathbf{R}^n).$$

It is not difficult to see that $\lambda_h \to \lambda$ if an only if

$$\sup\{|\lambda_h|(\Omega): h \in N\} < +\infty \qquad \text{and} \qquad \langle \lambda_h, u \rangle_\Omega \to \langle \lambda, u \rangle_\Omega \text{ for every } u \in C_c(\Omega; \mathbf{R}^n).$$

Moreover, if $\lambda_h \in \mathfrak{M}(\Omega)$ are positive measures and $\lambda_h \to \lambda$, then

$$\int_\Omega u \, \mathrm{d}\lambda \le \liminf_{h \to +\infty} \int_\Omega u \, \mathrm{d}\lambda_h \tag{2.3}$$

for every lower semicontinuous (shortly, l.s.c.) function $u: \Omega \to [0, +\infty]$, and

$$\int_C u \, \mathrm{d}\lambda \ge \limsup_{h \to +\infty} \int_C u \, \mathrm{d}\lambda_h \tag{2.4}$$

for every nonnegative $u \in C_0(\Omega)$ and every closed subset C of Ω.

Definition 2.1. We say that $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ is

(i) absolutely continuous with respect to μ (and we write $\lambda \ll \mu$) if

$$|\lambda|(B) = 0 \text{ whenever } B \in \mathcal{B} \text{ and } \mu(B) = 0;$$

(ii) singular with respect to μ (and we write $\lambda \perp \mu$) if

$$|\lambda|(\Omega - B) = 0 \text{ for a suitable } B \in \mathcal{B} \text{ with } \mu(B) = 0.$$

In the following, given $u \in L^1(\Omega); \mathbf{R}^n)$ we shall denote by $u \cdot \mu$ (or simply by u when no confusion is possible) the measure of $\mathfrak{M}(\Omega; \mathbf{R}^n)$ defined by

$$(u \cdot \mu)(B) = \int_B u \, \mathrm{d}\mu \qquad \text{for every Borel set } B;$$

moreover, if $u: \Omega \to \mathbf{R}$ is a bounded Borel function and $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ we denote by $u \cdot \lambda$ the measure of $\mathfrak{M}(\Omega; \mathbf{R}^n)$

$$(u \cdot \lambda)(B) = \int_B u \, \mathrm{d}\lambda \qquad \text{for every Borel set } B.$$

It is well-known that every measure $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ which is absolutely continuous with respect to μ is representable in the form $\lambda = u \cdot \mu$ for a suitable $u \in L^1(\Omega; \mathbf{R}^n)$; moreover, by the Lebesgue–Nikodym decomposition theorem, for every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ there exists a unique function $u \in L^1(\Omega; \mathbf{R}^n)$ and a unique measure $\lambda^s \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ such that

(i) $\lambda = u \cdot \mu + \lambda^s$;

(ii) λ^s is singular with respect to μ.

The function u is often indicated by $\mathrm{d}\lambda/\mathrm{d}\mu$.

When $\varphi: \Omega \times \mathbf{R}^n \to [0, +\infty]$ is a Borel function, with $\varphi(x, \cdot)$ positively 1-homogeneous for every $x \in \Omega$, it is well-known (see for instance Goffman and Serrin [11]) that for every $B \in \mathcal{B}$

and every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ the quantity

$$\int_B \varphi\left(x, \frac{d\lambda}{d\vartheta}\right) d\vartheta$$

does not depend on ϑ, when ϑ varies over all positive measures such that $|\lambda| \ll \vartheta$. Therefore, in the following, we shall denote the quantity above simply by

$$\int_B \varphi(x, \lambda).$$

For every subset B of Ω we denote by 1_B the function

$$1_B(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \in \Omega - B; \end{cases}$$

moreover, for every $x \in \Omega$ we indicate by δ_x the measure of $\mathfrak{M}(\Omega; \mathbf{R}^n)$

$$\delta_x(B) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \in \Omega - B. \end{cases}$$

Besides $\mathfrak{M}(\Omega; \mathbf{R}^n)$ the following spaces of measures will be considered:

$\mathfrak{M}^0(\Omega; \mathbf{R}^n)$ the spaces of all nonatomic measures of $\mathfrak{M}(\Omega; \mathbf{R}^n)$;

$\mathfrak{M}^\#(\Omega; \mathbf{R}^n)$ the space of all "purely atomic" measures of $\mathfrak{M}(\Omega; \mathbf{R}^n)$, that is the measures of the form

$$\lambda = \sum_{i=1}^{\infty} a_i \delta_{x_i}.$$

It is easy to see that for every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ there exist two unique measures $\lambda^0 \in \mathfrak{M}^0(\Omega; \mathbf{R}^n)$ and $\lambda^\# \in \mathfrak{M}^\#(\Omega; \mathbf{R}^n)$ (respectively called nonatomic part and atomic part of λ) such that $\lambda = \lambda^0 + \lambda^\#$. In other words we have

$$\mathfrak{M}(\Omega; \mathbf{R}^n) = \mathfrak{M}^0(\Omega; \mathbf{R}^n) \oplus \mathfrak{M}^\#(\Omega; \mathbf{R}^n).$$

For every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ we simply write $\lambda(x)$ instead of $\lambda(\{x\})$, and we denote by A_λ the set of all atoms of λ, that is

$$A_\lambda = \{x \in \Omega : \lambda(x) \neq 0\}.$$

In the following we shall denote by $\operatorname{cl} E$ the closure of a subset E of $\mathbf{R}^n$, and by $\operatorname{int} E$ its interior. For every $\rho > 0$, $B_\rho(s)$ will denote the ball $\{t \in \mathbf{R}^n : |t - s| < \rho\}$.

For every convex l.s.c. and proper function $f: \mathbf{R}^n \to]-\infty, +\infty]$ we define as usual the conjugate function f^* by

$$f^*(s) = \sup\{s \cdot w - f(w) : w \in \mathbf{R}^n\} \qquad (s \in \mathbf{R}^n).$$

The function $f^*: \mathbf{R}^n \to]-\infty, +\infty]$ is convex l.s.c. and proper, and we have $f^{**} = f$. We define also the recession function f^∞ of f by

$$f^\infty(s) = \lim_{t \to +\infty} \frac{f(s_0 + ts)}{t} \tag{2.5}$$

where s_0 is any point such that $f(s_0) < +\infty$. It is well-known that the definition above does not depend on s_0, and that the function f^∞ turns out to be convex, l.s.c., and positively 1-homogeneous on $\mathbf{R}^n$ (see for instance Laurent [13, proposition 6.8.3]).

In the sequel we deal with Borel functions $f: \Omega \times \mathbf{R}^n \to [0, +\infty]$ such that $f(x, \cdot)$ is convex, l.s.c., and proper for μ-a.e. $x \in \Omega$. For this kind of functions we define for every $(x, s) \in \Omega \times \mathbf{R}^n$

$$\varphi_f(x, s) = \sup\left\{u(x)s: u \in C_0(\Omega; \mathbf{R}^n), \int_\Omega f^*(x, u)\, d\mu < +\infty\right\}. \tag{2.6}$$

The function $\varphi_f(x, s)$ is l.s.c. in (x, s), convex and positively 1-homogeneous in s, and we have (see for instance Bouchitté & Valadier [3, proposition 7]):

$$\varphi_f(x, \cdot) \le f^\infty(x, \cdot) \qquad \text{for } \mu\text{-a.e. } x \in \Omega;$$

$$\varphi_f(x, \cdot) \ge f^\infty(x, \cdot) \qquad \text{for every } x \in \Omega \qquad \text{if } x \mapsto \operatorname{epi} f^*(x, \cdot) \text{ is l.s.c. on } \Omega.$$

In particular we have $\varphi_f = f^\infty$ when f does not depend on x.

For every function $g: \mathbf{R}^n \to [0, +\infty]$ we define

$$g^0(s) = \limsup_{t \to 0^+} \frac{g(ts)}{t} \qquad (s \in \mathbf{R}^n). \tag{2.7}$$

The following proposition holds.

Proposition 2.2. Let $g: \mathbf{R}^n \to [0, +\infty]$ be a l.s.c. function, with $g(0) = 0$, and such that

$$g(s_1 + s_2) \le g(s_1) + g(s_2) \qquad \text{for every } s_1, s_2 \in \mathbf{R}^n.$$

Then we have:

(i) the function $g^0: \mathbf{R}^n \to [0, +\infty]$ is convex, l.s.c., and positively 1-homogeneous;

(ii) $g^0(s) = \sup_{t>0}(g(ts)/t) = \lim_{t \to 0^+}(g(ts)/t)$ for every $s \in \mathbf{R}^n$.

Proof. Fix $s \in \mathbf{R}^n$ and set for every $r > 0$

$$\alpha(r) = \sup\left\{\frac{g(ts)}{t} : 0 < t \le r\right\}.$$

Clearly, the function α is increasing and satisfies

$$g^0(s) = \lim_{r \to 0^+} \alpha(r).$$

From the subadditivity of g we get for every $r > 0$

$$\alpha(2r) = \sup_{0 < t \le r} \frac{g(2ts)}{2t} \le \sup_{0 < t \le r} \frac{g(ts)}{t} = \alpha(r).$$

which implies that the function α is constant. Therefore

$$g^0(s) = \sup_{t>0} \frac{g(ts)}{t},$$

and so g^0 is positively 1-homogeneous, l.s.c., and subadditive on $\mathbf{R}^n$. Hence g^0 is convex, and (i) is proved. The proof of (ii) can be found for instance in Kuczma [12, theorem 3, p. 410]). ■

We remark that in general the subadditivity condition of g does not imply that $g(ts)/t)$ is decreasing in t, as shown by the functions $g(s) = |\sin s|$ and

$$g(s) = \begin{cases} 0 & \text{if } s \in \mathbf{Q} \\ 1 & \text{if } s \notin \mathbf{Q}. \end{cases}$$

In the following it will be useful to approximate g^0 from below by a sequence of convex and positively 1-homogeneous functions $g_\varepsilon: \mathbf{R}^n \to [0, +\infty]$ such that $g_\varepsilon(s) \le g(s)$ for every $s \in \mathbf{R}^n$ with $|s| \le \varepsilon$. More precisely, the following proposition holds.

PROPOSITION 2.3. Let $g: \mathbf{R}^n \to [0, +\infty]$ be as in proposition 2.2. For every $\varepsilon > 0$ define

$$C_\varepsilon = \{s^* \in \mathbf{R}^n: ss^* \le g(s) \text{ for every } |s| \le \varepsilon\};$$

$$g_\varepsilon(s) = \sup\{ss^*: s^* \in C_\varepsilon\}.$$

Then g_ε converges increasingly to g^0 as $\varepsilon \to 0$. Moreover, for every $s \neq 0$ we have

$$g_\varepsilon(s) = \inf\left\{\frac{g(ts)}{t} : \frac{\varepsilon}{2|s|} \le t \le \frac{\varepsilon}{|s|}\right\}. \tag{2.8}$$

Proof. Let us define

$$\varphi_\varepsilon(s) = \begin{cases} \inf\left\{\dfrac{g(ts)}{t} : \dfrac{\varepsilon}{2|s|} \le t \le \dfrac{\varepsilon}{|s|}\right\} & \text{if } s \neq 0 \\ 0 \quad \text{if } s = 0. \end{cases}$$

By using the subaddivity of g, it is easy to check that $\varphi_\in$ is convex, positively 1-homogeneous, and that

$$\varphi_\varepsilon(s) = \inf\left\{\frac{g(ts)}{t} : 0 < t < \frac{\varepsilon}{|s|}\right\} \qquad \text{for every } s \neq 0.$$

Thus φ_ε converges increasingly as $\varepsilon \to 0$, and by proposition 2.2

$$\lim_{\varepsilon \to 0} \varphi_\varepsilon(s) = \liminf_{t \to 0^+} \frac{g(ts)}{t} = g^0(s).$$

Hence, we have only to prove that the equality $g_\varepsilon \ldots, \varphi_\varepsilon$ holds for every $\varepsilon > 0$. Let us show first that φ_ε is l.s.c. Indeed, it is trivial at 0, and if $s_h \to s$ with $s \neq 0$, for a suitable $t_h \in [\varepsilon/2|s_h|, \varepsilon/|s_h|]$ we have

$$\liminf_{h \to \infty} \varphi_\varepsilon(s_h) \ge \liminf_{h \to \infty} \frac{g(t_h s_h)}{t_h}.$$

Possibly passing to subsequences, we may assume that $t_h \to t$ with $\varepsilon/2|s| \le t \le \varepsilon/|s|$. Then, by the semicontinuity of g

$$\liminf_{h\to\infty} \varphi_\varepsilon(s_n) \ge \frac{g(ts)}{t} \ge \varphi_\varepsilon(s).$$

Now, the easy computation of the polar function $(\varphi_\varepsilon)^*$ leads to

$$\varphi_\varepsilon^*(s^*) = \sup_{0<t|s|\le\varepsilon} ss^* - \frac{g(ts)}{t} = \sup_{t>0}\frac{1}{t}\left[\sup_{0<|w|\le\varepsilon} ws^* - g(w)\right]$$

which is nothing but the indicator function of the set C_ε. As $\varphi_\varepsilon = (\varphi_\varepsilon)^{**}$, we obtain $g_\varepsilon = \varphi_\varepsilon$, and so equality (2.8) is proved. ■

Given two function α, β: $\mathbf{R}^n \to [0, +\infty]$ we define the inf-convolution $\alpha\nabla\beta$ by

$$(\alpha\nabla\beta)(s) = \inf\{\alpha(t) + \beta(s-t)\colon t \in \mathbf{R}^n\}.$$

It is easy to see that

$$f\nabla f^\infty = f \qquad \text{for every } f\colon \mathbf{R}^n \to [0, +\infty] \text{ convex, l.s.c., proper,}$$

$$g\nabla g = g \qquad \text{for every } g\colon \mathbf{R}^n \to [0, +\infty] \text{ subadditive, with } g(0) = 0.$$

Moreover, the following proposition holds.

PROPOSITION 2.4. Let α, β: $\mathbf{R}^n \to [0, +\infty]$ be two convex, l.s.c., proper functions, with α such that

$$\lim_{|s|\to+\infty} \alpha(s) = +\infty. \tag{2.9}$$

Then we have

(i) $\alpha\nabla\beta$ is l.s.c. and $\alpha\nabla\beta = (\alpha^* + \beta^*)^*$;

(ii) $\alpha\nabla\beta_h \uparrow \alpha\nabla\beta$ for every sequence $\beta_h\colon \mathbf{R}^n \to [0, +\infty]$ of l.s.c. functions with $\beta_h \uparrow \beta$.

Proof. The proof of (i) can be found for instance in Laurent [13, théorème 6.5.4 and proposition 6.5.5]. Let us prove (ii), that is

$$\sup\{(\alpha\nabla\beta_h)(s)\colon h \in \mathbf{N}\} \ge (\alpha\nabla\beta)(s) \qquad \text{for every } s \in \mathbf{R}^n. \tag{2.10}$$

Fix $s \in \mathbf{R}^n$ such that the left-hand side of (2.10) is finite; then there exist $M > 0$ and a sequence (s_h) such that

$$\alpha(s_h) + \beta_h(s - s_h) \le (\alpha\nabla\beta_h)(s) + 1/h \le M \qquad \text{for every } h \in \mathbf{N}.$$

Condition (2.9) implies that (s_h) is bounded, so that we may suppose $s_h \to t$ for a suitable $t \in \mathbf{R}^n$. Therefore, for every $k \in \mathbf{N}$, by using the lower semicontinuity of α and β_k, we get

$$\begin{aligned}\alpha(t) + \beta_k(s-t) &\le \liminf_{h\to\infty}[\alpha(s_h) + \beta_k(s - s_h)]\\ &\le \liminf_{h\to\infty}[\alpha(s_h) + \beta_h(s - s_h)] \le \sup_{h\in\mathbf{N}}(\alpha\nabla\beta_h)(s),\end{aligned}$$

and so (ii) follows by taking first the supremum over all $k \in \mathbf{N}$ and then the infimum over all $t \in \mathbf{R}^n$. ■

Finally, by the following result we shall reduce the study of lower semicontinuity of a general functional to the simpler case of coercive functionals.

PROPOSITION 2.5. Let X be a Banach space with dual X', and let $F: X' \to [-\infty, +\infty]$ be a functional. Define for all $\varepsilon > 0$

$$F_\varepsilon(y) = F(y) + \varepsilon\|y\| \qquad (y \in X').$$

Then F is sequentially weakly* l.s.c. if and only if F_ε is sequentially weakly* l.s.c. for all $\varepsilon > 0$.

Proof. The only if part is trivial. Conversely, let $y_h \to y$ weakly* in X'; then, for a suitable $K > 0$

$$\sup\{\|y_h\|: h \in \mathbf{N}\} \le K.$$

Therefore, for every $\varepsilon > 0$

$$F(y) \le F_\varepsilon(y) \le \liminf_{h\to\infty} F_\varepsilon(y_h) \le \liminf_{h\to\infty} F(y_h) + \varepsilon K,$$

and so the conclusion follows by letting $\varepsilon \to 0$. ■

3. THE LOWER SEMICONTINUITY THEOREM

In this section we study the lower semicontinuity of functions of the form (1.5). Proposition 3.2 below shows that functionals of this form do not change if we replace f, φ, g by other "equivalent" functions.

LEMMA 3.1. Let B be an analytic subset of Ω (for instance $B \in \mathcal{B}$) such that

$$B \text{ is negligible for every finite positive nonatomic measure } \vartheta. \tag{3.1}$$

Then B is at most countable.

Proof. Assume by contradiction B is uncountable; then by corollary 3.5.2 of Rogers and Jayne [14] (to which we refer for the definition and properties of analytic sets, see also Cohn [7]), the set B contains a closed subset homeomorphic to $\{0, 1\}^{\mathbf{N}}$. Therefore, the standard nonatomic probability measure on $\{0, 1\}^{\mathbf{N}}$

$$\nu = \bigotimes_{h\in N} \{\tfrac{1}{2}\delta_0 + \tfrac{1}{2}\delta_1\}$$

provides by homeomorphism a nonatomic probability measure on B, which contradicts assumption (3.1). ■

PROPOSITION 3.2. Let $f_i, \varphi_i, g_i: \Omega \times \mathbf{R}^n \to [0, +\infty]$ be Borel functions ($i = 1, 2$), with $\varphi_i(x, \cdot)$ positively 1-homogeneous for every $x \in \Omega$. Then we have

$$\begin{aligned}\int_\Omega f_1\left(x, \frac{d\lambda}{d\mu}\right) d\mu + \int_{\Omega - A_\lambda} \varphi_1(x, \lambda^s) + \int_{A_\lambda} g_1(x, \lambda(x))\, d\# \\ = \int_\Omega f_2\left(x, \frac{d\lambda}{d\mu}\right) d\mu + \int_{\Omega - A_\lambda} \varphi_2(x, \lambda^s) + \int_{A_\lambda} g_2(x, \lambda(x))\, d\#\end{aligned} \tag{3.2}$$

for every $\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n)$, if and only if the following conditions hold:

(i) there exists $M \in \mathcal{B}$ with $\mu(M) = 0$ such that $f_1 = f_2$ on $(\Omega - M) \times \mathbf{R}^n$;

(ii) there exists a countable set $N \subset \Omega$ such that $\varphi_1 = \varphi_2$ on $(\Omega - N) \times \mathbf{R}^n$;

(iii) $g_1 = g_2$ on $\Omega \times \mathbf{R}^n$.

Proof. The if part is obvious. Let us assume now (3.2) holds; taking $\lambda = u \cdot \mu$ with $u \in L^1(\Omega; \mathbf{R}^n)$ we obtain immediately condition (i) (see for instance Buttazzo and Dal Maso [6]). Condition (iii) follows by taking $\lambda = s \cdot \delta_x$ with $x \in \Omega$ and $s \in \mathbf{R}^n$. Finally, in order to prove condition (ii), let us consider the set

$$B = \{x \in \Omega\colon \varphi_1(x, s) \neq \varphi_2(x, s) \text{ for some } s \in \mathbf{R}^n\}.$$

Since φ_1 and φ_2 are Borel functions, the set B is analytic, and so measurable for any Borel measure ϑ. Take now any positive $\vartheta \in \mathfrak{M}^0(\Omega)$; by (3.2) we get

$$\int_\Omega \varphi_1(x, u)\, \mathrm{d}\vartheta = \int_\Omega \varphi_2(x, u)\, \mathrm{d}\vartheta \qquad \text{for every } u \in L^1_\vartheta(\Omega; \mathbf{R}^n).$$

Hence, the same argument as before shows that B is ϑ-negligible, and then the conclusion follows from Lemma 3.1. ■

We can now state our lower semicontinuity result.

THEOREM 3.3. Let $f, \varphi, g\colon \Omega \times \mathbf{R}^n \to [0, +\infty]$ be Borel functions, and let f^∞, φ_f, g^0 be the functions defined by (2.5), (2.6), (2.7) respectively. Assume there exist $M \in B$ with $\mu(M) = 0$ and a countable set $N \subset \Omega$ such that:

(H_1) $f(x, \cdot)$ is convex and l.s.c. for every $x \in \Omega - M$;

(H_2) $f(x, u_0(x)) \in L^1(\Omega)$ for a suitable $u_0 \in L^1(\Omega; \mathbf{R}^n)$;

(H_3) $f^\infty = \varphi_f$ on $(\Omega - M) \times \mathbf{R}^n$;

(H_4) g is l.s.c. on $\Omega \times \mathbf{R}^n$;

(H_5) $g(x, s_1 + s_2) \leq g(x, s_1) + g(x, s_2)$ for every $x \in \Omega$ and $s_1, s_2 \in \mathbf{R}^n$;

(H_6) $g \leq \varphi_f$ on $\Omega \times \mathbf{R}^n$;

(H_7) $g^0 = \varphi = \varphi_f$ on $(\Omega - N) \times \mathbf{R}^n$.

Then the functional

$$F(\lambda) = \int_\Omega f\left(x, \frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_\lambda} \varphi(x, \lambda^s) + \int_{A_\lambda} g(x, \lambda(x))\, \mathrm{d}\#$$

is sequentially l.s.c. on $\mathfrak{M}(\Omega; \mathbf{R}^n)$ with respect to the weak* convergence.

Remark 3.4. We stress the fact that the functional F appearing in theorem 3.3 is l.s.c. without being convex; the convexity of F occurs only when $g(x, \cdot)$ is convex and positively 1-homogeneous for all $x \in \Omega$.

Remark 3.5. When all the integrands do not depend on the variable x, theorem 3.3 gives the lower semicontinuity of the functional

$$\int_\Omega f\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_\lambda} f^\infty(\lambda^s) + \int_{A_\lambda} g(\lambda(x))\, \mathrm{d}\#$$

under the only conditions:

(a) f is convex, l.s.c., and proper on $\mathbf{R}^n$.

(b) g is l.s.c. and subadditive on $\mathbf{R}^n$, and verifies $g^0 = f^\infty$.

For instance, the functional

$$F(\lambda) = \begin{cases} \int_\Omega |u|^2 \, d\mu + \#(\operatorname{spt} \nu) & \text{if } \lambda = u \cdot \mu + \nu,\ u \in L^1(\Omega; \mathbf{R}^n),\ \nu \in \mathfrak{M}^\#(\Omega; \mathbf{R}^n) \\ +\infty & \text{otherwise} \end{cases}$$

is sequentially weakly* l.s.c. on $\mathfrak{M}(\Omega; \mathbf{R}^n)$, as well as the functional

$$F(\lambda) = |\lambda|(\Omega - A_\lambda) + \sum_{x \in A_\lambda} |\sin(\lambda(x))|.$$

In the proof of theorem 3.3 we shall use the following lemmas.

LEMMA 3.6. Let k be a positive integer, and let

$$\mathfrak{M}_k = \{\lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n) : \#(\operatorname{spt} \lambda) \le k\}.$$

Then $\mathfrak{M}_k$ is sequentially weakly* closed. Moreover, for every l.s.c. function $g: \Omega \times \mathbf{R}^n \to [0, +\infty]$ satisfying the subadditivity condition (H_5) of theorem 3.3, the functional

$$G(\lambda) = \int_\Omega g(x, \lambda(x)) \, d\#(x)$$

is sequentially weakly* l.s.c. on $\mathfrak{M}_k$.

Proof. Let $\lambda_h \to \lambda$ with $\lambda_h \in \mathfrak{M}_k$; then

$$\lambda_n = \sum_{i=1}^{k} s_{i,h} \delta_{x_{i,h}} \qquad \text{for suitable } x_{i,h} \in \Omega,\ s_{i,h} \in \mathbf{R}^n.$$

Moreover,

$$\sup\left\{ \sum_{i=1}^{k} |s_{i,h}| : h \in \mathbf{N} \right\} < +\infty.$$

Denote by I the set of integers $i \in \{1, \dots, k\}$ such that $(x_{i,h})_{j \in \mathbf{N}}$ has a subsequence converging in Ω; possibly passing to subsequences, we may assume that for every $i \in I$ it is $x_{i,h} \to x_i$ and $s_{i,h} \to s_i$ for suitable $x_i \in \Omega$ and $s_i \in \mathbf{R}^n$. On the other hand, if $i \notin I$ it is easy to see that

$$s_{i,h} \delta_{x_{i,h}} \to 0 \qquad \text{weakly* in } \mathfrak{M}(\Omega; \mathbf{R}^n).$$

Therefore we have

$$\lambda = \sum_{i \in I} s_i \delta_{x_i}$$

and so $\lambda \in \mathfrak{M}_k$. We prove now the lower semicontinuity of G. By the lower semicontinuity of $g(x, s)$, and recalling that $g \ge 0$, we have

$$\liminf_{h \to \infty} G(\lambda_h) \ge \liminf_{h \to \infty} \sum_{i \in I} g(x_{i,h}, s_{i,h}) \ge \sum_{i \in I} g(x_i, s_i),$$

and, by using the subadditivity of $g(x, \cdot)$,

$$\sum_{i \in I} g(x_i, s_i) \geq \int_\Omega g(x, \lambda(x))\, \mathrm{d}\#(x),$$

so that the proof is achieved. ■

LEMMA 3.7. Let $f: \Omega \times \mathbf{R}^n \to [0, +\infty]$ be a Borel function satisfying conditions (H_1) and (H_2) of theorem 3.3, and let $C(x)$ be a multifunction from Ω to the closed convex subsets of $\mathbf{R}^n$ such that

(i) C is l.s.c. (that is the set $\{x \in \Omega : C(x) \cap U \neq \emptyset\}$ is open whenever $U \subset \mathbf{R}^n$ is open);
(ii) $C(x) \subset \operatorname{dom} \varphi_f^*(x, \cdot)$ for every $x \in \Omega$;
(iii) there exists $r > 0$ such that $[s \in \mathbf{R}^n : |s| < r\} \subset C(x)$ for every $x \in \Omega$.

Then, for every $\lambda \in \mathfrak{M}(\Omega, \mathbf{R}^n)$ we have

$$\sup\left\{\int_\Omega u\, \mathrm{d}\lambda - \int_\Omega f^*(x, u)\, \mathrm{d}\mu : u \in C_0(\Omega; \mathbf{R}^n),\ u(x) \in C(x)\ \forall x \in \Omega\right\}$$
$$= \int_\Omega [f(x, \cdot)\, \nabla h(x, \cdot)]\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_\Omega h(x, \lambda^s) \tag{3.3}$$

where $h(x, s) = \sup\{ss^* : s^* \in C(x)\}$.

Proof. Let us consider the measure $m = \mu + |\lambda^s|$ and

$$j(x, s) = f^*(x, s) \frac{\mathrm{d}\mu}{\mathrm{d}m}(x) \qquad (x, s) \in \Omega \times \mathbf{R}^n;$$

$$J(u) = \int_\Omega j(x, u)\, \mathrm{d}m \qquad u \in L_m^\infty(\Omega; \mathbf{R}^n);$$

$$H = \{u \in C_0(\Omega; \mathbf{R}^n) : u(x) \in C(x) \quad \forall x \in \Omega, J(u) < +\infty\};$$

$$\Gamma(x) = \mathrm{cl}\{u(x) : u \in H\} \qquad x \in \Omega.$$

We prove now that $\Gamma(x) = C(x)$ for every $x \in \Omega$. The inclusion $\Gamma(x) \subset C(x)$ is trivial because $C(x)$ is closed. Since $C(x)$ is convex and with nonempty interior, in order to prove the opposite inclusion, it is enough to show that $\operatorname{int} C(x) \subset \Gamma(x)$. Let $s \in \operatorname{int} C(x)$ and let $\rho > 0$ be such that $B_\rho(s) \subset C(x)$; since the multifunction C is l.s.c., by lemma 15 of Bouchitté and Valadier [3] there exists a neighbourhood V of x such that

$$B_{\rho/2}(s) \subset C(y) \qquad \text{for every } y \in V.$$

By condition (ii) and by the convexity of J, we obtain

$$s \in \operatorname{int}[\operatorname{dom} \varphi_f^*(x, \cdot)] = \operatorname{int}[\mathrm{cl}\{u(x) : u \in C_0(\Omega; \mathbf{R}^n), J(u) < +\infty\}]$$
$$= \operatorname{int}\{u(x) : u \in C_0(\Omega; \mathbf{R}^n), J(u) < +\infty\}.$$

Then, there exists $u \in C_0(\Omega; \mathbf{R}^n)$ with $J(u) < +\infty$ such that $u(x) = s$, so that for a suitable neighbourhood W of x, with $W \subset V$, we have

$$u(y) \in B_{\rho/2}(s) \text{ hence } u(y) \in C(y) \qquad \text{for every } y \in W.$$

By using the convexity of J and C, and a standard truncation argument, we may construct a function $v \in C_0(\Omega; \mathbf{R}^n)$ such that

$$v(x) = s, \qquad v(y) \in C(y) \text{ for every } y \in \Omega, \qquad J(v) < +\infty.$$

Hence $v \in H$, and so $s \in \Gamma(x)$.

Now, we use theorem 2 of Bouchitté and Valadier [3], and we get

$$(J + I_H)^*(w) = \int_\Omega k(x, w)\, \mathrm{d}m \qquad \text{for every } w \in L^1_m(\Omega; \mathbf{R}^n)$$

where $k(x, s) = [j^*(x, \cdot)\, \nabla h(x, \cdot)]^{**}(s)$ and I_H is the indicator function of H (i.e. $I_H = 0$ on H and $I_H = +\infty$ otherwise). An easy computation shows that

$$j^*(x, \cdot)\, \nabla h(x, \cdot) = \begin{cases} f(x, \cdot)\, \nabla h(x, \cdot) & \text{for } \mu\text{-a.e. } x \in \Omega \\ h(x, \cdot) & \text{for } \lambda^s\text{-a.e. } x \in \Omega, \end{cases}$$

and taking into account the coerciveness of $h(x, \cdot)$ and proposition 2.4(i), we finally obtain

$$(J + I_H)^* \left(\frac{\mathrm{d}\lambda}{\mathrm{d}m}\right) = \int_\Omega [f(x, \cdot)\, \nabla h(x, \cdot)] \left(\frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_\Omega h(x, \lambda^s)$$

which is nothing else but (3.3). ■

Proof of theorem 3.3. By proposition 2.5 we may assume there exists $c > 0$ such that

$$F(\lambda) \geq c|\lambda|(\Omega) \qquad \text{for every } \lambda \in \mathfrak{M}(\Omega; \mathbf{R}^n),$$

hence

$$f(x, s) \geq c|s|, \qquad \varphi(x, s) \geq c|s|, \qquad g(x, s) \geq c|s| \qquad \text{for every } (x, s) \in \Omega \times \mathbf{R}^n. \tag{3.4}$$

Set for every $\varepsilon > 0$ and every $(x, s) \in \mathbf{R}^n$

$$C_\varepsilon(x) = \{s^* \in \mathbf{R}^n : ss^* \leq g(x, s) \text{ for every } s \in \mathbf{R}^n \text{ with } |s| \leq \varepsilon\}$$

$$g_\varepsilon(x, s) = \sup\{ss^* : s^* \in C_\varepsilon(x)\}.$$

As in the proof of proposition 2.3, we can show that g_ε is l.s.c. on $\Omega \times \mathbf{R}^n$; hence by theorem 17 of Bouchitté and Valadier [4], $C_\varepsilon(x)$ is a l.s.c. multifunction.

Let now (λ_h) be a sequence converging to λ weakly* in $\mathfrak{M}(\Omega; \mathbf{R}^n)$, and let $\varepsilon > 0$ be fixed. Set

$$A_h = \{x \in \Omega : |\lambda_h(x)| \geq \varepsilon\}, \qquad \lambda_{1,h} = \lambda_h \cdot 1_{\Omega - A_h}, \qquad \lambda_{2,h} = \lambda_h \cdot 1_{A_h}.$$

Possibly passing to subsequences, we may assume there exist $\lambda_1, \lambda_2 \in \mathfrak{M}(\Omega; \mathbf{R}^n)$ such that $\lambda_1 + \lambda_2 = \lambda$ and

$$\lambda_{1,h} \to \lambda_1 \qquad \text{and} \qquad \lambda_{2,h} \to \lambda_2 \qquad \text{weakly* in } M(\Omega; \mathbf{R}^n).$$

By lemma 3.6 we have that $\lambda_2 \in \mathfrak{M}^\#(\Omega; \mathbf{R}^n)$ and

$$\begin{aligned} \liminf_{h \to \infty} F(\lambda_{2,h}) &= \liminf_{h \to \infty} \int_\Omega g(x, \lambda_{2,h}(x))\, \mathrm{d}\#(x) \\ &\geq \int_\Omega g(x, \lambda_2(x))\, \mathrm{d}\#(x) = F(\lambda_2). \end{aligned} \tag{3.5}$$

Let now $u \in C_0(\Omega; \mathbf{R}^n)$ be such that $u(x) \in C_\varepsilon(x)$ for every $x \in \Omega$; since

$$C_\varepsilon(x) \subset \operatorname{dom}(g^0)^*(x, \cdot) \subset \operatorname{dom}(\varphi_F)^*(x, \cdot)$$

we have

$$s \cdot u(x) \le \varphi_f(x, s) \qquad \text{for every } (x, s) \in \Omega \times \mathbf{R}^n$$

$$s \cdot u(x) \le g(x, s) \qquad \text{for every } x \in \Omega \text{ and every } s \in \mathbf{R}^n \text{ with } |s| \le \varepsilon.$$

Then, since $|\lambda_{1,h}(x)| \le \varepsilon$ for every $x \in \Omega$,

$$\begin{aligned}
&\int_\Omega f\left(x, \frac{\mathrm{d}\lambda_{1,h}}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_{\lambda 1,h}} \varphi(x, \lambda^s_{1,h}) + \int_\Omega g(x, \lambda_{1,h}(x))\, \mathrm{d}\# \\
&\quad \ge \int_\Omega \left[u \cdot \frac{\mathrm{d}\lambda_{1,h}}{\mathrm{d}\mu} - f^*(x, u)\right] \mathrm{d}\mu + \int_{\Omega - A_{\lambda_{1,h}}} u\, \mathrm{d}\lambda^s_{1,h} + \int_\Omega u(x) \cdot \lambda_{1,h}(x)\, \mathrm{d}\# \\
&\quad = \int_\Omega u\, \mathrm{d}\lambda_{1,h} - \int_\Omega f^*(x, u)\, \mathrm{d}\mu,
\end{aligned}$$

and so

$$\liminf_{h \to \infty} F(\lambda_{1,h}) \ge \int_\Omega u\, \mathrm{d}\lambda_1 - \int_\Omega f^*(x, u)\, \mathrm{d}\mu.$$

Taking the supremum with respect to u when $u(x) \in C_\varepsilon(x)$ for every $x \in \Omega$, by lemma 3.7 we obtain

$$\liminf_{h \to \infty} F(\lambda_{1,h}) \ge \int_\Omega [f(x, \cdot)\, \nabla g_\varepsilon(x, \cdot)]\left(\frac{\mathrm{d}\lambda_1}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_\Omega g_\varepsilon(x, \lambda^s_1).$$

Therefore, by letting $\varepsilon \to 0$ and taking into account propositions 2.3 and 2.4,

$$\liminf_{h \to \infty} F(\lambda_{1,h}) \ge \int_\Omega [f(x, \cdot)\, \nabla g^0(x, \cdot)]\left(\frac{\mathrm{d}\lambda_1}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_\Omega g^0(x, \lambda^s_1).$$

By hypotheses (H_3) and (H_7) we have

$$f(x, \cdot)\, \nabla g^0(x, \cdot)\, \nabla f^\infty(x, \cdot) = f(x, \cdot) \qquad \text{for } \mu\text{-a.e. } x \in \Omega,$$

and so, recalling that $g^0 \ge g$, we may write

$$\begin{aligned}
\liminf_{h \to \infty} F(\lambda_{1,h}) &\ge \int_\Omega f\left(x, \frac{\mathrm{d}\lambda_1}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_{\lambda_1}} \varphi(x, \lambda^s_1) + \int_\Omega g^0(x, \lambda_1(x))\, \mathrm{d}\# \\
&\ge \int_\Omega f\left(x, \frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_\lambda} \varphi(x, \lambda^s) + \int_\Omega g(x, \lambda_1(x))\, \mathrm{d}\#. \qquad (3.6)
\end{aligned}$$

Putting together (3.5) and (3.6), recalling that $\lambda_{1,h} \perp \lambda_{1,h}$, and using the subadditivity condition (H_5), we finally obtain

$$\begin{aligned}
\liminf_{h \to \infty} F(\lambda_h) &= \liminf_{h \to \infty} [F(\lambda_{1,h}) + F(\lambda_{2,h})] \ge \liminf_{h \to \infty} F(\lambda_{1,h}) + \liminf_{h \to \infty} F(\lambda_{2,h}) \\
&\ge \int_\Omega f\left(x, \frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_\lambda} \varphi(x, \lambda^s) + \int_\Omega [g(x, \lambda_1(x)) + g(x, \lambda_2(x))]\, \mathrm{d}\# \\
&\ge \int_\Omega f\left(x, \frac{\mathrm{d}\lambda}{\mathrm{d}\mu}\right) \mathrm{d}\mu + \int_{\Omega - A_\lambda} \varphi(x, \lambda^s) + \int_\Omega g(x, \lambda(x))\, \mathrm{d}\# = F(\lambda). \quad \blacksquare
\end{aligned}$$

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REFERENCES

1. AMBROSIO L. & BUTTAZZO G., Weak lower semicontinuous envelope of functions defined on a space of measures, *Ann. Mat. pura appl.* **150**, 311–340 (1988).
2. BOUCHITTÉ G., Représentation intégrale de fonctionnelles convexes sur un espace de mesures, *Ann. Univ. Ferrara* **33**, 113–156 (1987).
3. BOUCHITTÉ G. & VALADIER M., Integral representation of convex functionals on a space of measures. *J. funct. Analysis* **80**, 398–420 (1988).
4. BOUCHITTÉ G. & VALADIER M., Multifonctions s.c.i. et régularisée s.c.i. essentielle. Fonctions de mesure dans le cas sous linéaire, Proc. "Congrès Franco-Québécois d'Analyse non Linéaire Appliquée", Perpignan, June 22–26, 1987 (to appear).
5. BUTTAZZO G., Semicontinuity, Relaxation and Integral Representation in the Calculus of Variations, *Pitman Research Notes in Mathematics* **207**. Longman, Harlow (1989).
6. BUTTAZZO G. & DAL MASO G., On Nemyckii operators and integral representation of local functionals, *Rc. Mat.* **3**, 491–509 (1983).
7. COHN D. L., *Measure Theory*. Birkhäuser, Basel (1980).
8. DE GIORGI E., Free discontinuity problems in calculus of variations, Proc. Int. Meeting in J. L. Lions's honour, Paris, June 6–10, 1988 (to appear).
9. DE GIORGI E. & AMBROSIO L., Un nuovo tipo di funzionale del calcolo delle variazioni. *Atti Acad. naz. Lincei Rc.* (to appear).
10. DE GIORI E., AMBROSIO L. & BUTTAZZO G., Integral representation and relaxation for functionals defined on measures. *Atti Accad. naz. Lincei Rc.* **81**, 7–13 (1987).
11. GOFFMAN C. & SERRIN J., Sublinear functions of measures and variational integrals. *Duke math J.* **31**, 159–178 (1964).
12. KUCZMA M., *An Introduction to the Theory of Functional Equations and Inequalities*. Panstwowe Wydawnictwo Naukowe, Warszawa (1985).
13. LAURENT P. J., *Approximation et Optimisation*. Hermann, Paris (1972).
14. ROGERS C. A. & JAYNE J. E., K-analytic sets, in *Analytic Sets* (Edited by C. A. ROGERS *et al.*). Academic Press, New York (1980).